

SONALIKA SINGH

+91 8434756189 | singhsonalika5@gmail.com | [LinkedIn](#) | [GitHub](#) | [Kaggle](#) | [Portfolio](#)

Professional Summary

Aspiring **data scientist** with a strong mathematical foundation from IIT Madras and hands-on experience across the **ML lifecycle** — data collection, **EDA**, model development (**classification, regression, forecasting**), and stakeholder-facing visualization. Skilled in **Python, SQL**, and deep learning frameworks with growing exposure to **MLOps** tooling.

Education

Master of Mathematics , Indian Institute of Technology Madras	7.43 CGPA	(2023–2025)
Bachelor of Science (Mathematics) , Patliputra University, Patna	62.2%	(2018–2021)
Senior Secondary , Bihar School Examination Board	68%	(2016–2018)
Secondary , Central Board of Secondary Education	9.4 CGPA	(2015–2016)

Experience

Data Annotator & Content Moderator (Freelance), Innodata (Remote) Dec 2025 – Present

- Annotated and validated large-scale text, image, audio, and video datasets used to train machine learning models, maintaining an average QA quality score of **3.12** (against a target of **2.00**) across **6+** months of continuous work.
- Ensured data consistency and compliance with annotation guidelines across multimodal datasets, directly improving downstream model performance.

Data Analyst Intern, ERNET India, IIT Madras Research Park Jul 2024 – Jun 2025

- Collected and cleaned **180+** days of weather and signal data for a **1.12 km** FSO link in Kohima; performed exploratory data analysis (EDA) in Python to identify outage causes and quantify downtime.
- Built **10+** Python visualization tools and KPI dashboards to monitor link performance and categorize outages, translating findings into actionable reliability recommendations for stakeholders.
- Developed and evaluated groundwater **forecasting** models (**MLR, RFR, ANN, GRU, LSTM**) on **2.5K+** data points, achieving **98%** accuracy for sustainable planning use cases.

Winter Intern – Applications of ML Techniques, E&ICT Academy, IIT Guwahati Dec 2024 – Jan 2025

- Designed, trained, and evaluated **5+** deep learning models (FFNNs, CNNs) for **classification** tasks using TensorFlow/Keras on MNIST and CIFAR-10, boosting accuracy by **15%**.
- Fine-tuned VGG and ResNet on custom datasets, benchmarking performance using **6+** evaluation techniques (Accuracy, Precision, Recall, F1, ROC-AUC, Confusion Matrix).

Projects

Multilingual Language Classifier

[GitHub](#)

Built an end-to-end MLOps pipeline classifying **27** Roman-script languages and continent groups on **10,000+** samples using character-level TF-IDF, TruncatedSVD, and Naive Bayes/LDA/QDA, achieving **96.8%** language and **89.4%** continent accuracy; deployed via Streamlit.

Sequence Modeling for Language Decoding

[GitHub](#)

Built and optimized Seq2Seq transliteration models (**RNN, LSTM, GRU**) with attention mechanisms and beam search decoding, achieving **82.23%** character-level test accuracy via W&B Bayesian hyperparameter sweeps.

Music Recommender System

[GitHub](#)

Built a content-based recommender preprocessing **1M+** song lyrics with NLTK, using TF-IDF (**5,000** features) and cosine similarity to deliver real-time **top-5** recommendations via a Streamlit app with Spotify API integration.

Multi-label EHR Classification (ML Challenge)

[GitHub](#)

Built a **4-layer** deep learning model (LeakyReLU, Dropout) for multi-label classification on **200K+** EHR embeddings, achieving a Micro F₂ score of **0.8183** through threshold tuning and data engineering.

Skills

Languages: Python, SQL

Domains: Natural Language Processing, Computer Vision, Large Language Models, Dimensionality Reduction

Frameworks & Tools: Scikit-learn, TensorFlow, PyTorch, NumPy, Pandas, NLTK, Matplotlib, Seaborn, Power BI, Tableau, Git, MLflow, DVC, Docker, WandB, Grafana

Relevant Coursework & Certifications

Core Coursework: Foundations of Machine Learning, Introduction to Deep Learning, Data Structures & Algorithms, Data Analytics Laboratory, Linear Algebra, Probability, Applied Statistics, Numerical Optimization, Multivariable Calculus

Certifications: **OCI Data Science Professional** (Oracle), **Data Science Job Simulation** (Lloyds Banking Group), Data Analytics (Deloitte), Intro to AI (IITM Summer School), Data Science (Udemy), Communication Skills (TCS iON), Power BI Workshop

Achievements & Activities

Academic: Secured **AIR 417** in IIT-JAM (Mathematics); qualified JEE (2018, 2019) and GATE Data Science (2025).

Leadership: Volunteer, Forays 2024 (Math Fest, IIT Madras) – managed event operations and participant coordination.

Sports: Represented IIT Madras hostel teams in Cricket & Football; reached **4th** position in cricket tournaments and demonstrated teamwork, adaptability, and continuous improvement.